

Effect of Modified ZnO on Electrical Properties of PP/SEBS Nanocomposites for HVDC Cables

Jun-Wei Zha,^{1,2} Jun-Fu Wang,¹ Si-Jiao Wang,^{1,2} Qingqing Qin,¹ and Zhi-Min Dang^{1,2}

¹School of Chemistry and Biological Engineering

University of Science and Technology Beijing, Beijing 100083, P. R. China

²Department of Electrical Engineering

Tsinghua University, Beijing 100084, P. R. China

ABSTRACT

Polypropylene (PP) has the potential to be used as insulation for high voltage direct current cables due to its high melting point, excellent recyclability and electrical insulation properties. The PP/SEBS composites filled with surface-modified ZnO (m-ZnO) nanoparticles were fabricated through melt blending. Scanning electron microscopy (SEM) showed that the m-ZnO nanoparticles were well dispersed in the PP/SEBS matrix. With doping of 0.5 wt% m-ZnO, the DC breakdown strength, space charge suppression and mechanical properties of the PP/SEBS were remarkably improved, and their DC conductivity was found to be lowered. Moreover, the incorporation of m-ZnO has little effect on the melting and crystallization process as well as dielectric properties of the PP/SEBS. Thus, the PP/SEBS/m-ZnO composites have shown promise to be used as insulation for recyclable HVDC cables.

Index Terms —polypropylene, ZnO nanoparticles, HVDC cable, thermoplastic elastomer, space charge

1 INTRODUCTION

WITH the rapid development of power industry, the high voltage direct current (HVDC) transmission technology has been generally considered to be an important component of the future smart grid and global energy interconnection. The extruded cables as crucial equipment of HVDC transmission system have wide prospect for application.

Polypropylene (PP) is a semi-crystalline thermoplastic polymer and has two kinds of crystalline and amorphous phases. Compared to traditional cross-linked polyethylene (XLPE), PP owns higher melting temperature, which could easily satisfy the requirement for higher voltage and larger transmission power. Importantly, the crosslinking process is unnecessary for PP cables, which will improve the production as well as reduce the cost. Besides, it is well known that PP is recoverable, which can be used as eco-friendly materials. Among the five conventional plastics, the production of PP is generally ranked third, which has low-cost advantage. Therefore, PP can be considered as an ideal insulating material for HVDC cables, which could meet the economic and environmental demands. Besides, PP has good thermal properties and high mechanical strength. The isotactic PP has a melting point of up to 160 °C and can be used under long-term operation temperature of 100-120 °C. It is a potential

environmental friendly material to be used in high voltage direct current (HVDC) cables [1-3]. However, due to its large modulus, PP has poor toughness and impact performance, and brittle fracture at low temperature.

Generally, the mechanical properties of PP could be improved by incorporation of some kinds of elastomers, such as EPDM, POE and SEBS [4-7]. SEBS as one new type of thermoplastic elastomer can be used at 130 °C. Besides, SEBS has good heat resistance and good thermal properties, and it can be recycled, which is also in line with the concept of environmental protection [8-11]. Importantly, PP has good compatibility with the hydrogenated polybutadiene chain in the elastomer SEBS. It is also an ideal material to improve the mechanical properties of PP.

It is known that the accumulation of space charge is one key problem for the HVDC cables. It was reported that the doping of nano-sized particles could effectively suppress the accumulation of space charge in the polymer insulation. Nano zinc oxide particles (ZnO) can be used as the biological antibacterial agents, UV protective agents, and information storage materials. Some scholars have added nano ZnO into the PP to study their optical properties and mechanical properties [12, 13]. However, there have been few reports on suppression of space charge by introduction of ZnO into PP blends.

In this paper, the PP/SEBS blends and the PP/SEBS/ZnO composites were prepared. The thermal and mechanical

properties, dielectric properties, DC breakdown strength and conductivity as well as the space charge behaviors of the prepared PP/SEBS/ZnO composites were investigated. It was observed that the introduction of SEBS in PP resulted in more accumulation of space charge. The space charge could be remarkably suppressed by the doping of ZnO into the PP/SEBS blends. In addition, the PP/SEBS/ZnO composites showed the enhanced mechanical flexibility. These results indicate that the PP/SEBS/ZnO composites have potential for HVDC cable application.

2 EXPERIMENTAL PROCEDURES

2.1 SAMPLES PREPARATION

Isotactic PP was purchased from Yanshan Petrochemical Company, Beijing. Thermoplastic elastomer SEBS was supplied from Taiwan rubber (Nantong Industrial Co., Ltd.). ZnO nanoparticles were provided by Beijing DK nano Technology Co. Ltd. The PP, elastomer SEBS and modified ZnO were blended in a twin-screw extruder for 20 min at 190 °C. The content of ZnO nanoparticles was 0.2, 0.5, 1 and 2%. The pellets were hot-pressed at 190 °C for 20 min using a plate vulcanizer under a pressure of 15 MPa, and then the samples were cooled to room temperature for 5 min. The composition and weight ratio of each type of specimen are shown in Table 1.

Table 1. The prepared samples with different proportions.

Designation	PP (wt%)	SEBS (wt%)	ZnO (wt%)
PP/SEBS	65	35	0
PP/SEBS/0.2%ZnO	65	35	0.2
PP/SEBS/0.5%ZnO	65	35	0.5
PP/SEBS/1.0%ZnO	65	35	1
PP/SEBS/2.0%ZnO	65	35	2

2.2 DIFFERENTIAL SCANNING CALORIMETRY (DSC) MEASUREMENT

The PP/SEBS/ZnO was placed in one aluminum crucible, and the crucible with a cap was placed in a pressure device. Then the sample mould was prepared for DSC testing. In the presence of nitrogen, the samples were heated from 30 to 200 °C and maintained at 200 °C for 2 min in order to eliminate the thermal history. Then the temperature was cooled down from 200 to 30 °C and maintained at 30 °C for 2 min. Afterwards, it was heated from 30 to 200 °C at a rate of 10 °C/min and maintained at 200 °C for 2 min, followed by the final natural cooling.

2.3 MECHANICAL PROPERTY TESTING

Tensile test was performed on universal testing machine (CMT4304) produced by Meitesi Industry System Co. Ltd. Dog-bone shaped specimens with the gauge length of 25 mm and the width of 4 mm were used. Their mechanical properties were evaluated by the typical stress-strain curve and elongation at break. Every five specimens were tested and their average value was recorded.

2.4 SPACE CHARGE MEASUREMENT

The space charge accumulation in the insulating material is a key factor restricting the development of HVDC cable. It was found that the accumulation of space charge would cause the distortion of local electric field and even partial discharge, resulting in the failure of insulation [14]. Therefore, the suppression of space charge has become one great concern for the HVDC insulation materials. It was reported that doping of nanoparticles in the polymer matrix can effectively suppress the accumulation of space charge and improve the breakdown field strength [15,16]. The space charge measurement was performed with a pulsed electro-acoustic (PEA) system. All the samples were under 60 kV/mm DC electrical field for 30 min. An electric pulse with an amplitude of 600 V and duration of 5 ns was applied to the samples and the space charge distributions during polarization process were recorded.

2.5 DC BREAKDOWN TESTING

High-voltage breakdown device with spherical electrodes locally produced by Tsinghua University was used. The plate samples with thickness of 150 µm prepared after vulcanization were placed between two sphere electrodes and immersed in transformer oil to prevent surface flashover. A ramp rate of 2 kV/s was applied until the sample broke down. 11 points were tested for each sample. The experimental results were analyzed by Weibull distribution.

2.6 DIELECTRIC PROPERTIES MEASUREMENT

Dielectric properties of the samples with thickness of 1 mm and the area of 1 cm² were carried out through an impedance analyzer (Agilent 4294 A). After the upper and lower sides of the sample were plated with copper electrodes, the sample was tested at the frequency ranging from 10³ Hz to 10⁶ Hz at room temperature.

2.7 CONDUCTIVITY MEASUREMENT

Conductive current and electrical conductivity were also important criteria for evaluating the insulation performance of insulation materials. The increased electrical conductivity will increase the current loss in the material. The samples with thickness of 0.12 mm was tested under 50 kV/mm for 300 s. The diameter of the electrode used was 2 cm. The current density of the sample was calculated by the formula:

$$J=I/S.....(1)$$

where J is the current density, I is the current, S is the area of sample and the conductivity of the sample was calculated as follows:

$$\sigma=J/E.....(2)$$

where σ is the conductivity and E is electrical field.

3 RESULTS AND DISCUSSIONS

3.1 MICROSTRUCTURAL CHARACTERIZATION OF THE PP/SEBS/ZnO COMPOSITES

As shown in Figure 1, it can be seen that the ZnO particles was evenly dispersed in the PP/SEBS bulk and there were almost no agglomerations. The ZnO particles are uniform with

a diameter of about 50 nm. It was because that the ZnO fillers after modification by KH550 could give rise to good adhesion effect with the polymer matrix. In this case, it was considered that the surface free energy of the nanoparticles was reduced [17]. And the good compatibility between nanoparticles and polymer could make them show excellent mechanical and electrical properties [18,19].

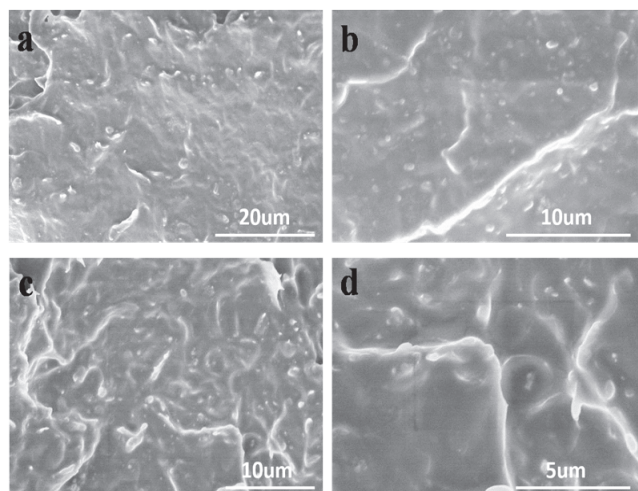


Figure 1. SEM images of the PP/SEBS/ZnO composites with the addition of ZnO (a) 0.2%, (b) 0.5%, (c) 1% and (d) 2%.

3.2 MELTING AND CRYSTALLIZATION CURVES OF THE PP/SEBS/ZnO COMPOSITES

Figures 2a and 2b show the melting and crystallization curves of the PP/SEBS blends with different concentration of ZnO, respectively. With gradually increasing the content of ZnO from 0 to 2%, it was found that the addition of ZnO has little effect on the melting point of the PP/SEBS, which is around 161 °C (Figure 2a). As shown in Figure 2b, with the addition of 0.2% ZnO, the crystallization temperature of the PP/SEBS/ZnO composites is almost the same as that of PP/SEBS blend. That is, the incorporation of small amount of ZnO (0.2, 0.5%) almost has no influence on the melting and crystallization process of the PP/SEBS, which is generally considered to be of great importance for the HVDC insulation.

3.3 BREAKDOWN STRENGTH OF THE PP/SEBS/ZnO COMPOSITES

The Weibull distribution of DC breakdown strength of the PP/SEBS and PP/SEBS/ZnO composites with surface-modified ZnO (0.2, 0.5, 1, and 2%) was shown in Figure 3. With increasing the ZnO content, it can be seen that the breakdown strength of PP/SEBS/ZnO composites was firstly increased and then decreased. When the content of ZnO is 0.5%, the value of breakdown strength is found to be the largest. Besides the good compatibility between ZnO and the matrix, the addition of ZnO will introduce deep traps, which could effectively capture the carriers, thus resulting in the increased breakdown strength. However, with doping of more ZnO particles, it will possibly lead to some agglomerations, which is equivalent to the introduction of some impurities in

the insulating matrix. The ionization of impurities will increase the concentration of free electrons. It is easier for the electronic migration, resulting in the reduced breakdown strength. The breakdown strength of samples with different ZnO nanoparticles loading is shown in Table 2.

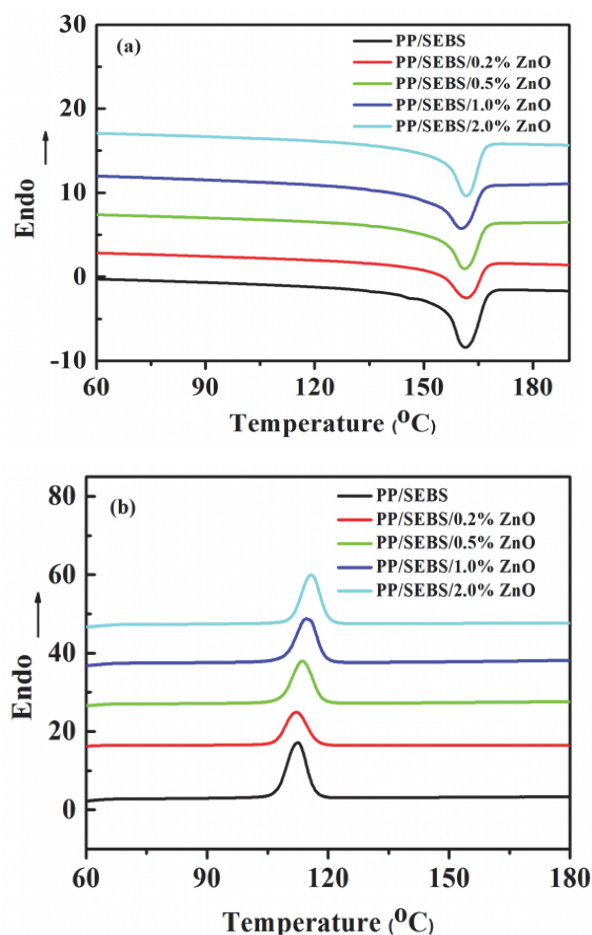


Figure 2. DSC curves of the PP/SEBS/ZnO composites during (a) melting process and (b) crystallization process.

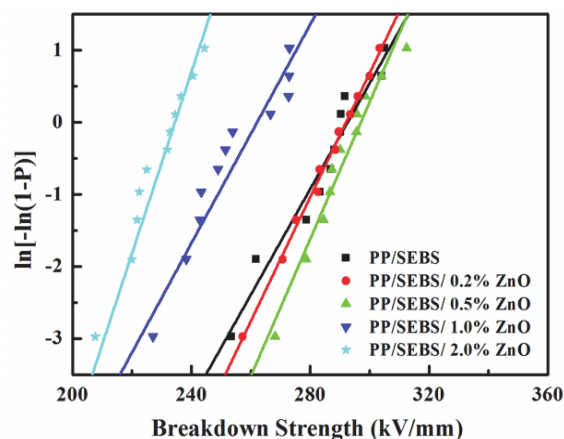


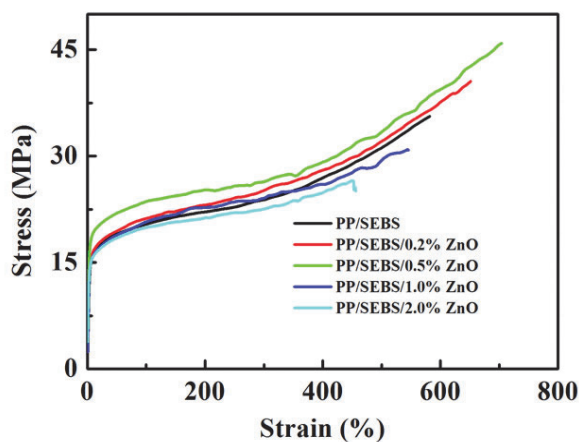
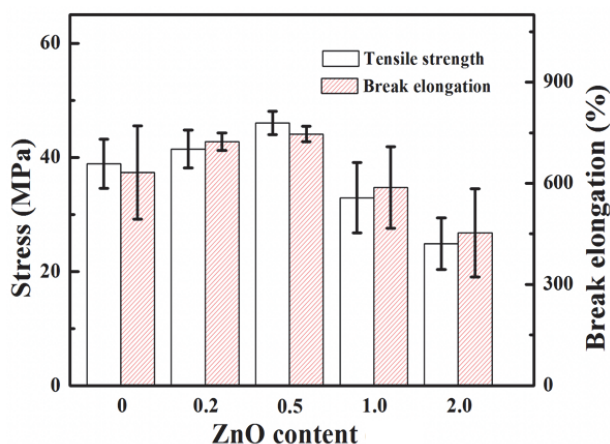
Figure 3. Weibull distribution of DC breakdown strength of PP/SEBS/ZnO nanocomposites.

Table 2. Breakdown strength of the samples.

ZnO(wt%)	0	0.2	0.5	1.0	2.0
Breakdown Strength (kV/mm)	293	292	298	261	232

3.4 MECHANICAL PROPERTY OF THE PP/SEBS/ZnO COMPOSITES

Figure 4 shows typical stress-strain curves of the PP/SEBS and PP/SEBS/ZnO composites with 0.2, 0.5, 1 and 2% surface-modified ZnO. The corresponding tensile strength and elongation at break were shown in Figure 5. It can be seen that the PP/SEBS/ZnO composites with 0.5% ZnO shows the largest tensile strength and elongation at break. Compared to the PP/SEBS, the addition of 0.5% ZnO makes the enhanced tensile strength up to 46.2 ± 1.7 MPa, which is increased by 18%. And its elongation at break was increased by 24.7%. The enhancement of mechanical properties was attributed to the large amount of interface production by doping of ZnO with PP/SEBS, which had different properties compared to pure polymer matrices [20-23]. The organic component of the interface modifier KH550 could be well fused with the PP, which will ensure good compatibility and interfacial adhesion between the nanoparticles and polymer matrix.

**Figure 4.** Tensile stress-strain curves of the PP/SEBS/ZnO composites.**Figure 5.** Tensile histogram of the PP/SEBS/ZnO nanocomposites

3.5 SPACE CHARGE BEHAVIORS OF THE PP/SEBS/ZnO COMPOSITES

The accumulation of space charge is considered to be the key factor for the development of HVDC cable insulation material. It was found that the accumulation of space charge in the HVDC electric field could cause the distortion of local electric field or even the partial discharge, ultimately resulting in the failure of insulation [14]. Therefore, the suppression of space charge has been focused on for the insulation materials. Some research has found that the doping nanoparticles in the polymer matrix can effectively suppress the accumulation of space charge and improve the breakdown field strength [15, 16]. The space charge distributions of the PP/SEBS and PP/SEBS/ZnO composites with 0.2, 0.5, 1, and 2% surface-modified ZnO are shown in Figure 6. And their corresponding electrical field distribution is shown in Figure 7.

As shown in Figure 6a, with the addition of the elastomer SEBS in the isotactic PP, a large amount of heterogeneous space charge was found to be injected into the two electrodes. And the injection depth increased with increasing the time, especially at the vicinity of the cathode. Figure 7a shows the corresponding electric field distribution. It can be seen that the accumulation of space charge results in the distortion of electric field. The distortion field strength after polarization is up to 9.0 kV/mm, which will probably affect the insulation performance of HVDC cables during their long-term operation. With the incorporation of ZnO (0.5 %) into the PP/SEBS as shown in Figure 6b, it is helpful for the suppression of the heterogeneous charge. That is because the doping of ZnO into the PP/SEBS matrix will introduce the deep traps, which can effectively capture the carriers and reduce the migration of carriers within the materials. With further increasing the ZnO content, some accumulation of space charge was found within the material and the distortion of electric field appeared.

For more addition of ZnO, it is easy for the formation of the paths of carriers, which will lead to the electronic migration through tunneling effect in the transition between adjacent traps. Thus, the accumulation of space charge and the distortion rate of field strength were increased. For the composites with 1 and 2% ZnO as shown in Figure 6d-e and Figure 7d-e, it can be seen that their space charge behavior and field distortion were almost the same. In this case, some agglomerations of ZnO possibly make the interaction between the ZnO and polymer matrix weakened. And the depth of some traps will become shallower. During the process from the deep to shallow traps, the partial charges will easily escape from the traps, so the injection of space charge was reduced. It is generally considered that the accumulation of space charge was related to the trapping, detrapping, and transporting characteristics of the charge carriers inside the material. As previously reported, the doping of nanoparticles into the polymer could introduce a large number of traps at the interface region, which will greatly affect the accumulation of space charge [24,25].

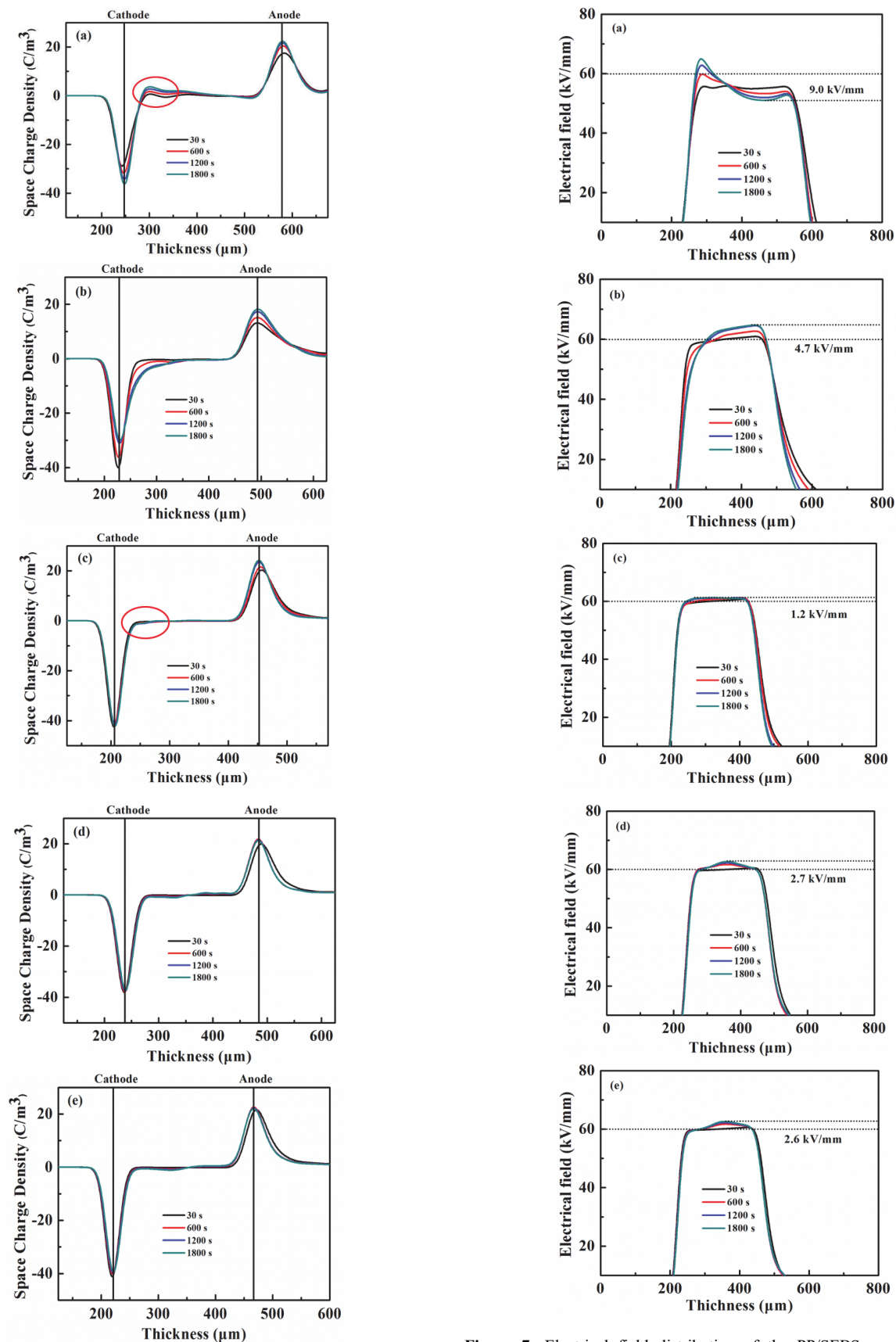


Figure 6. Space charge distribution of the PP/SEBS composites with addition of ZnO (a) 0 %, (b) 0.2 %, (c) 0.5 %, (d) 1 %, (e) 2 % during polarization subjected to DC electric field of 60 kV/mm for 30 min.

Figure 7. Electrical field distribution of the PP/SEBS composites with addition of ZnO (a) 0 %; (b) 0.2 %; (c) 0.5 %; (d) 1 %; (e) 2 % during polarization subjected to DC electric field of 60 kV/mm for 30 min.

3.6 DIELECTRIC PROPERTIES OF THE PP/SEBS/ZNO COMPOSITES

The relative dielectric permittivity and dielectric loss of the PP/SEBS and PP/SEBS/ZnO composites with 0.2, 0.5, 1 and 2% surface-modified ZnO were shown in Figure 8. It can be seen that their relative dielectric permittivity was firstly decreased and then increased. With the addition of ZnO (0.5%), it was found that the relative dielectric permittivity was lowered to be 2.67, which was obviously lower than that of the PP/SEBS. It was consistent with the previous results where the permittivity of composites was lower than that of the unfilled polymers [26-28]. It was attributed to the hindrance in the movement of polymer chains, which is resulted from the entanglements of the polymer chains or the interaction of the polymer with the nanoparticles [29-31]. When the content of ZnO was up to 1 and 2%, their dielectric permittivity was increases, which is due to the interfacial polarization between the ZnO fillers and polymer matrix [32]. It can be seen from Figure 8b that the addition of ZnO has less effect on the dielectric loss of the PP/SEBS matrix, and most of their value is lower than 0.02 with the frequency ranging from 10^3 to 10^6 Hz.

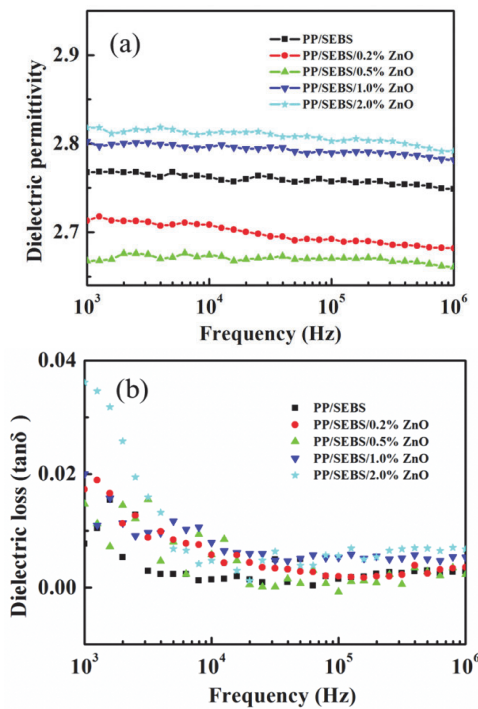


Figure 8. Dependence of (a) relative permittivity and (b) dielectric loss of the PP/SEBS/ZnO composite on frequency from 10^3 to 10^6 Hz.

3.7 CONDUCTIVITY CHARACTERISTICS OF PP/SEBS/ZNO COMPOSITES

The conductivity, current and current density of PP/SEBS and PP/SEBS/ZnO composites with 0.2, 0.5, 1 and 2% surface-modified ZnO were shown in Figure 9. It can be seen that the current, current density and conductivity of PP/SEBS/ZnO were also firstly decreased and then increased. When the content of ZnO was 0.5%, the conductivity was 0.7×10^{-15} S/m, which was lower than that of PP/SEBS. With increasing the

content of ZnO to 1 and 2%, the conductivity was increased over that of PP/SEBS. And similar results have been reported in the MgO/LDPE composites [33]. This is because that the nanoparticles are polarized under DC electric field and the potential well always originates at the interface between the particles and polymer matrix. The potential wells will capture the charge carriers and reduce their transport rate, which will give rise to the decreased conductivity. Moreover, when the content of nanoparticles is lower, it will produce a number of potential wells. However, for adding the larger amounts of nanoparticles, the matrix adhesion will be decreased due to the agglomeration of particles, and, resulting the increase of conductivity [34,35].

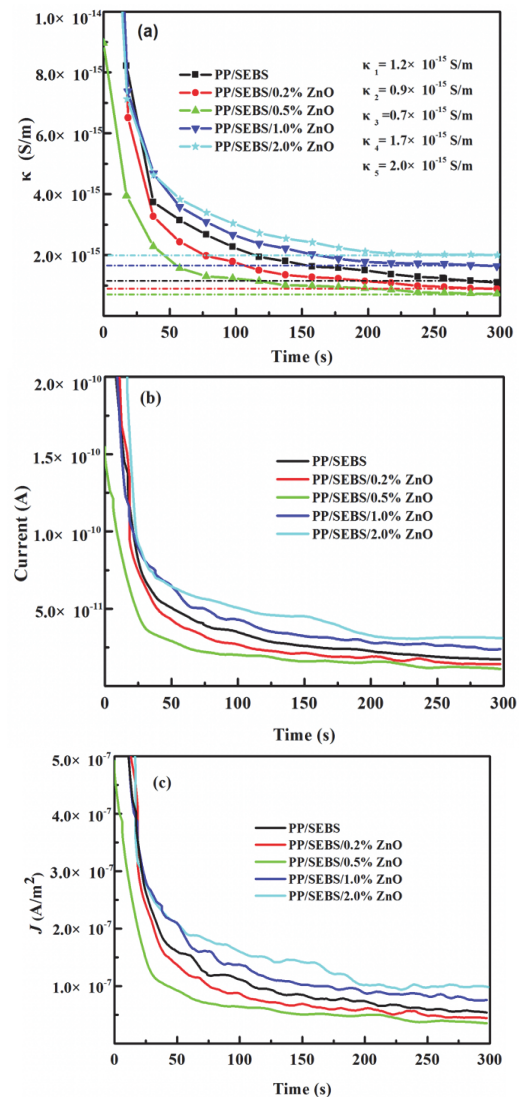


Figure 9. The (a) conductivity, (b) current and (c) current density curves of the PP/SEBS and PP/SEBS/ZnO composite.

4 CONCLUSIONS

The PP/SEBS/ZnO composites with ZnO surface-modified by the silane coupling agent (KH550) were prepared by melt blending method. We can see from the SEM images that the modified ZnO were well dispersed in the PP/SEBS matrix.

And the addition of ZnO has little effect on the melting and crystallization of PP/SEBS. The DC breakdown strength and conductivity, dielectric and space charge behaviors as well as mechanical properties of the PP/SEBS/ZnO composites were mainly discussed. The tensile strength with elongation at break of the composites were improved with the doping of 0.5% ZnO. The optimal content of ZnO for improving the DC electrical properties of the composites was set as 0.5%, and they exhibited higher breakdown strength, lower conductivity and remarkable suppression of space charge. It was found that the composites had relatively stable permittivity and low dielectric loss, which indicating their application among a broad frequency range. The results show that the PP/SEBS composites with 0.5% ZnO have promising applications to be used as the insulation materials for the recyclable HVDC cables.

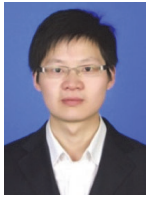
ACKNOWLEDGEMENTS

The authors gratefully acknowledge support from the National Basic Research Program of China (973 Program, No. 2014CB239501), the National Natural Science Foundation of China (Nos. 51425201, 51622701). Beijing Nova program (Z181100006218006), the Fundamental Research Funds for the Central Universities (No. FRF-TP-16-001C1). Z. M. D. also thanks the Postgraduate Education Reform Project of Tsinghua University (53812000216) for financial support.

REFERENCES

- [1] A. van der Wal, J. J. Mulder, J. Oderkerk, R. J. Gaymans, "Polypropylene-rubber blends: The effect of the matrix properties on the impact behavior", *Polymer*, vol. 39, no. 26, pp. 6781-6787, 1998.
- [2] E. Helal, E. David, M. Fr  chette and N. R. Demarquette, "Thermoplastic elastomer nanocomposites with controlled nanoparticles dispersion for HV insulation: correlation between rheological, thermal, electrical and dielectric properties", *Europ. Polym. J.*, vol. 94, pp. 68-86, 2017.
- [3] E. Helal, N. R. Demarquette, E. David, M. Fr  chette, "Evaluation of dielectric behavior of polyethylene/thermoplastic elastomer blends containing zinc oxide (ZnO) nanoparticles for high voltage insulation", in *Proc. IEEE Int. Conf. Electr. Insul.*, 2016, pp. 592-595.
- [4] K. Jokannessor, J. Antonischki, A. Ericssonk, "The development of an extruded HVDC cable system", in *Jicable*, 1999, pp. 201-205.
- [5] X. Hong, G. Nie, Z. Lin, and J. Rong, "Structure and properties of PPO/PP blends compatibilized by triblock copolymer SEBS and SEPS", *Polym.-Plast. Technol. Eng.*, vol. 51, no. 10, pp. 971-976, 2012.
- [6] O. D. Jazani, A. Arefazar, S. H. Jafari, P. R. Mohammad, S. R. Mohammad, and T. Alireza, "SEBS-g-MAH as a reactive compatibilizer precursor for PP/PTT/SEBS ternary blends: morphology and mechanical properties", *Polym.-Plast. Technol. Eng.*, vol. 52, no. 2, pp. 206-212, 2013.
- [7] B. X. Du, H. Xu, J. Li, Z. Li, "Space charge behaviors of PP/POE/ZnO nanocomposites for HVDC cables", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 23, no. 5, pp. 3165-3174, 2016.
- [8] A. Cordova, "High molecular weight styrenic block copolymer hot melt adhesive for toy articles", US Patent 6797766, Sep. 28, 2004.
- [9] S. V. Ranade, R. E. Richard, M. N. Helmus, "Styrenic block copolymers for biomaterial and drug delivery applications", *Acta Biomaterialia*, vol. 1, no. 1, pp. 137-144, 2005.
- [10] I. Luzinov, D. Julthongpipit, P. D. Bloom, V. V. Sheares, V. V. Tsukruk, "Bilayer nanocomposite molecular coatings from elastomeric/rigid polymers, fabrication, morphology and micromechanical properties", *Macromol. Symp.*, vol. 167, pp. 227-242, 2001.
- [11] K. Sipkens, "SEBS-based compounds", *Medical Device Technol.*, vol. 11, pp. 8-10, 2000.
- [12] R. Yang, Y. Li, J. Yu, "Photo-stabilization of linear low density polyethylene by inorganic nano-particles", *Polym. Degrad. Stability*, vol. 88, pp. 168-174, 2005.
- [13] X. Zhang, M. Liu, G. Li, "Nano-ZnO modified PE", *Plastic*, vol. 33, pp. 9-15, 2004.
- [14] G. C. Montanari, "Bringing an insulation to failure: the role of space charge", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 18, no. 2, pp. 339-364, 2011.
- [15] S. Ju, H. Zhang, M. Chen, C. Zhang, X. Chen, Z. Zhang, "Improved electrical insulating properties of LDPE based nanocomposite: Effect of surface modification of magnesium nanoparticles", *Compos. A: Appl. Sci. Manufact.*, vol. 66, pp. 183-192, 2014.
- [16] X. Dou, X. Liu, Y. Zhang, H. Feng, J. F. Chen, and S. Du, "Improved dielectric strength of barium titanate-polyvinylidene fluoride nanocomposite", *Appl. Phys. Lett.*, vol. 95, pp. 132904-1-3, 2009.
- [17] Y. Zhou, J. He, J. Hu, B. Dang, "Surface-modified MgO nanoparticle enhances the mechanical and direct-current electrical characteristics of polypropylene/polyolefin elastomer nanodielectrics", *J. Appl. Polym. Sci.*, vol. 133, pp. 42863-1-10, 2016.
- [18] X. Huang, F. Liu, P. Jiang, "Effect of nanoparticle surface treatment on morphology, electrical and water treeing behavior of LLDPE composites", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 17, no. 6, pp. 1697-1074, 2010.
- [19] B. P. Singh, D. Singh, R. B. Mathur, T. L. Dhami, "Influence of surface modified MWCNTs on the mechanical, electrical and thermal properties of polyimide nanocomposites", *Nanoscale Res. Lett.*, vol. 3, pp. 444-453, 2008.
- [20] R. Y. Hong, H. P. Fu, Y. J. Zhang, L. Liu, J. Wang, H. Z. Li, Y. Zheng, "Surface-modified silica nanoparticles for reinforcement of PMMA", *J. Appl. Polym. Sci.*, vol. 105, no. 4, pp. 2176-2184, 2007.
- [21] C. L. Wu, M. Q. Zhang, M. Z. Rong, K. Friedrich, "Silica nanoparticles filled polypropylene: effects of particle surface treatment, matrix ductility and particle species on mechanical performance of the composites", *Compos. Sci. Technol.*, vol. 65, pp. 635-645, 2005.
- [22] S. H. Lee, M. Kontopoulou, C. B. Park, "Effect of nanosilica on the co-continuous morphology of polypropylene/polyolefin elastomer blends", *Polymer*, vol. 51, no. 5, pp. 1147-1155, 2010.
- [23] A. Galeski, "Strength and toughness of crystalline polymer systems", *Progr. Polym. Sci.*, vol. 28, no. 12, pp. 1643-1699, 2003.
- [24] F. Tian, Q. Q. Lei, X. Wang, and Y. Wang, "Effect of deep trapping states on space charge suppression in polyethylene/ZnO nanocomposite", *Appl. Phys. Lett.*, vol. 99, pp. 142903, 2011.
- [25] B. Han, X. Wang, Z. Sun, J. Yang, and Q. Lei, "Space charge suppression induced by deep traps in polyethylene/zeolite nanocomposite", *Appl. Phys. Lett.*, vol. 102, pp. 012902, 2013.
- [26] Y. Cao, P. C. Irwin, K. Younsi, "The Future of nanodielectrics in the electrical power industry", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 11, no. 5, pp. 797-807, 2004.
- [27] J. K. Nelson and Y. Hu, "Nanocomposite dielectrics-properties and implications", *J. Phys. D: Appl. Phys.*, vol. 38, no. 2, pp. 213-222, 2005.
- [28] S. Singha and M. J. Thomas, "Dielectric properties of epoxy nanocomposites", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 15, no. 1, pp. 12-23, 2008.
- [29] M. Roy, J. K. Nelson, R. K. MacCrone, L. S. Schadler, "Candidate mechanisms controlling the electrical characteristics of silica/XLPE nanodielectrics", *J. Mater. Sci.*, vol. 42, no. 11, pp. 3789-3799, 2007.
- [30] S. Singha and M. J. Thomas, "Influence of filler loading on dielectric properties of epoxy-ZnO nanocomposites", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 16, no. 2, pp. 531-542, 2009.
- [31] S. Li, W. Wang, S. Yu, H. Sun, "Influence of hydrostatic pressure on dielectric properties of polyethylene/aluminum oxide nanocomposites", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 21, no. 2, pp. 519-528, 2014.
- [32] L. Zhang, Y. Zhou, M. Huang, Y. Sha, J. Tian, Q. Ye, "Effect of nanoparticle surface modification on charge transport characteristics in XLPE/SiO₂ nanocomposites", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 21, no. 2, pp. 424-433, 2014.
- [33] K. Ishimoto, E. Kanegae, Y. Ohki, et al. "Superiority of dielectric properties of LDPE/MgO nanocomposites over microcomposites", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 16, no. 6, pp. 1735-1742, 2009.
- [34] T. Takada, Y. Hayase, Y. Tanaka, T. Okamoto, "Space charge trapping in electrical potential well caused by permanent and induced dipoles for LDPE/MgO nanocomposite", *IEEE Trans. Dielectr. Electr. Insul.*, vol. 15, no. 1, pp. 152-160, 2008.

- [35] J. I. Hong, L. S. Schadler, and R. W. Siegel, "Rescaled electrical properties of ZnO/low density polyethylene nanocomposites", Appl. Phys. Lett., vol. 82, no. 12, pp. 1956-1958, 2003.



J. W. Zha received Ph.D. degree in materials science and engineering in 2010 from Beijing University of Chemical Technology. He studied in University of Southampton, as a Joint-Ph.D student sponsored by China Scholarship Council from 2009 to 2010. He currently works in the Department of Polymer Science and Engineering, University of Science & Technology Beijing. He has published more than 70 journal papers.



J. Wang received the B.Sc degree in chemistry from Langfang Teachers University in 2015. Then she received the Master degree in 2017 from University of Science & Technology Beijing.



S. J. Wang received her Ph.D degree from University of Science & Technology Beijing. She is currently a postdoctoral researcher in Department of Electrical Engineering, Tsinghua University.



Q. Qin received her Bachelor degree in chemistry from Xuchang University in 2016. She currently studies in the Department of Polymer Science and Engineering, University of Science & Technology Beijing for Master degree.



Z. M. Dang received his Ph.D degree in electrical engineering from Xi'an Jiaotong University in 2001. He is currently a Professor in Department of Electrical Engineering, Tsinghua University. He is also a member of IEEE on Dielectric and Electrical Insulation. His present research interests are electrical functional materials. He has published more than 180 journal papers with a current citation record of over 5000 times and an H-index of 40.